

# Exploratory Analysis of Breast Cancer Diagnostic Features

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## Declaration of Original Work

I declare that this project is my own original work. All sources of information and assistance have been acknowledged. This submission has not been previously submitted for assessment at this or any other institution.

Signature:  Date: 04/06/2026

## Abstract

Breast cancer is among the most prevalent malignancies worldwide, and accurate discrimination between malignant and benign tumours is critical for clinical decision-making. This study presents an exploratory data analysis of the Breast Cancer Wisconsin Diagnostic Dataset (UCI Machine Learning Repository, accession ID 17), comprising 569 patient samples characterised by 30 numeric cellular measurements derived from digitised fine-needle aspirate images. Group means were computed for malignant ( $n=212$ ) and benign ( $n=357$ ) classes across all analytical features, and features were ranked by standardized effect size (Cohen's  $d$ ) rather than raw absolute mean difference, so that features measured on very different scales could be compared fairly. The five most discriminatory features were worst concave points, worst perimeter, mean concave points, worst radius, and mean perimeter (all  $d > 2.2$ , indicating very large effect sizes) — descriptors of nuclear boundary irregularity and cell size. Malignant samples exhibited substantially higher values across all five features, consistent with known characteristics of nuclear pleomorphism and uncontrolled tumour cell growth. These findings demonstrate the utility of standardized statistical comparison for identifying diagnostically informative biomarkers and provide a foundation for more advanced predictive modelling.

## 1. Introduction

Breast cancer is the most commonly diagnosed cancer in women globally and a leading cause of cancer-related mortality. According to the World Health Organization, over 2.3 million new cases are diagnosed annually, underscoring the critical need for reliable and early diagnostic tools. Histological and cytological examination of tissue samples remains central to diagnosis, but manual interpretation is time-consuming and subject to inter-observer variability. Computational approaches that systematically analyse quantitative cellular measurements offer the potential to improve diagnostic consistency and reproducibility.

The advent of digitised medical imaging and open-access datasets has created new opportunities for data-driven analysis in oncology. Machine learning and statistical methods can be applied to large collections of cell morphology measurements to identify patterns that distinguish malignant from benign tissue — patterns that may not be immediately apparent through visual inspection alone. Such approaches complement clinical judgement and have demonstrated strong predictive performance in a range of cancer diagnostic tasks.

This study aims to identify which cellular measurements differ most markedly between malignant and benign breast tumour samples by performing a structured exploratory analysis of the Breast Cancer Wisconsin Diagnostic Dataset. The objective is to rank the top five most discriminatory features using a standardized effect-size comparison, and to communicate findings through reproducible, version-controlled code and scientific visualisation.

## 2. Literature Context

The use of machine learning in cancer diagnostics has been extensively validated in the literature. Kourou et al. (2015) reviewed the application of machine learning methods to cancer prognosis and prediction, demonstrating that supervised classifiers applied to clinical and genomic data consistently achieve high accuracy. The Breast Cancer Wisconsin dataset itself has become a benchmark in this field; Mangasarian et al. (1995) demonstrated that linear programming classifiers trained on cell

nucleus measurements from fine-needle aspirates could achieve over 97% diagnostic accuracy. These studies establish that quantitative morphological features carry strong discriminatory signal for malignancy classification.

The importance of open-access, reproducible datasets in advancing biomedical research is well recognised. Wilkinson et al. (2016) articulated the FAIR principles — that scientific data should be Findable, Accessible, Interoperable, and Reusable — which underpin modern data sharing standards. The UCI Machine Learning Repository, which hosts the dataset used in this study, exemplifies these principles by providing freely available, well-documented datasets that enable independent replication of analyses. Reproducibility through version-controlled workflows and open code is increasingly regarded as a minimum standard in computational biology (Sandve et al., 2013).

### 3. Objectives

1. To retrieve the Breast Cancer Wisconsin Diagnostic Dataset from the UCI Machine Learning Repository and organise it within a version-controlled repository.
2. To clean the dataset by inspecting for missing values, excluding non-analytical columns from statistical comparison, and saving a processed copy for downstream analysis.
3. To identify the five cellular features that show the greatest standardized effect size (Cohen's d) between malignant and benign diagnostic classes.
4. To visualise the top five discriminatory features as an effect-size bar chart and the single most discriminatory feature as a boxplot, with self-contained captions reporting the statistics needed to interpret each figure independently.
5. To interpret the findings in the context of known breast cancer cell biology and discuss limitations and directions for future work.

### 4. Dataset and Tools Overview

The Breast Cancer Wisconsin (Diagnostic) Dataset was obtained from the UCI Machine Learning Repository (Wolberg et al., 1995). The dataset contains 569 patient records, of which 212 are labelled malignant (M) and 357 benign (B), representing a class distribution of approximately 37% and 63% respectively. Each record includes 30 numeric features derived from digitised images of fine-needle aspirate (FNA) biopsies. These features describe the characteristics of cell nuclei and are computed as the mean, standard error, and worst (largest) value for ten base measurements: radius, texture, perimeter, area, smoothness, compactness, concavity, concave points, symmetry, and fractal dimension.

| Tool / Library | Version | Purpose                       |
|----------------|---------|-------------------------------|
| Python         | 3.8+    | Programming language          |
| pandas         | 2.x     | Data loading and manipulation |
| numpy          | 1.x     | Numerical computation         |
| matplotlib     | 3.x     | Figure generation             |

|           |     |                                     |
|-----------|-----|-------------------------------------|
| seaborn   | 0.x | Statistical visualisation (boxplot) |
| reportlab | 4.x | PDF report generation               |

## 5. Methodology and Computational Workflow

The analysis followed a six-stage pipeline: Data Acquisition → Inspection → Cleaning → Group Comparison → Visualisation → Reporting. This sequential workflow was implemented as a single reproducible Python script (*scripts/analysis.py*) and executed in its entirety to produce all outputs.

**Data Acquisition and Inspection.** The dataset was loaded from a local CSV file using pandas, with column names assigned from the known WDBC schema since the raw file has no header row. The first analytical step was a class balance check using `value_counts()` on the diagnosis column, which confirmed 357 benign and 212 malignant samples. A missing value audit using `isnull().sum()` confirmed no missing entries across any of the 30 feature columns, eliminating the need for imputation.

**Cleaning.** The id column was excluded from all group-mean and effect-size calculations, as it is a non-analytical record identifier rather than a biological measurement. This exclusion is deliberate: retaining id in a feature-ranking step can cause it to appear as an artificially strong 'discriminator' purely because of how records happen to be ordered in the file, not because it carries any real diagnostic signal. A `dropna()` call was included as a precautionary measure to handle any edge-case null values, though none were present. The cleaned DataFrame was saved to `data/processed/breast_cancer_clean.csv` to maintain a clear separation between raw and processed data.

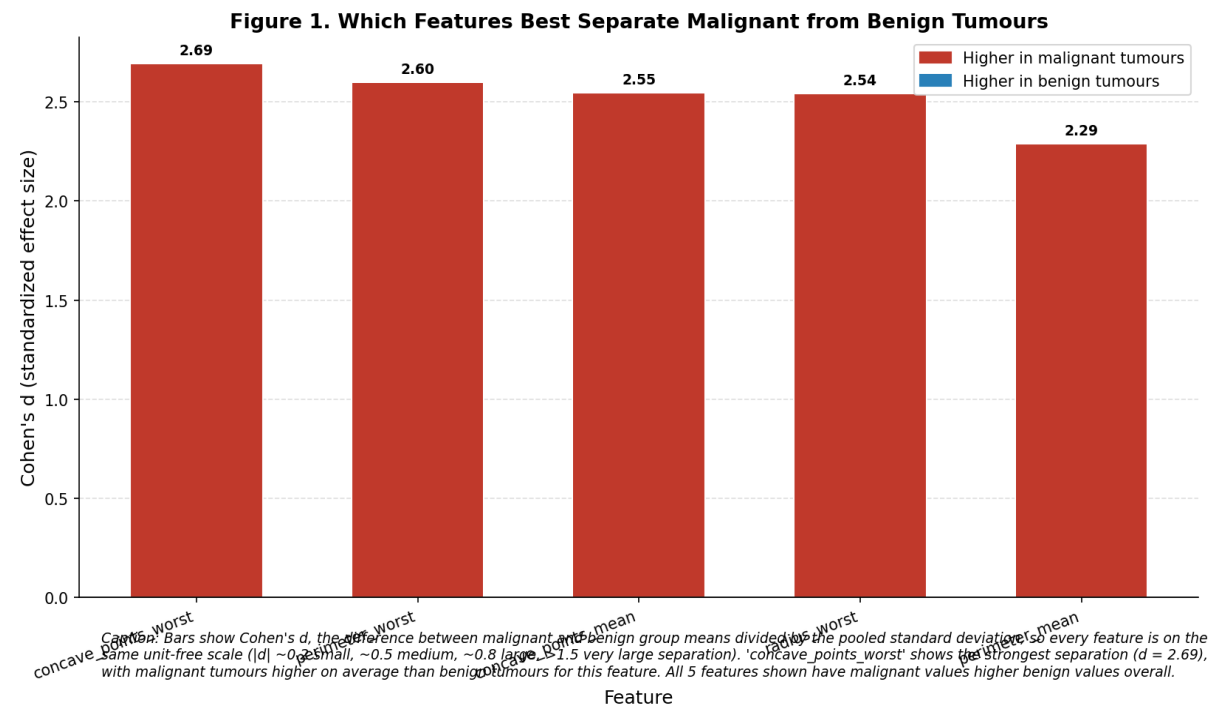
**Group Comparison and Ranking.** The dataset was split into malignant and benign subsets using the analytical feature columns only. Rather than ranking features by raw absolute mean difference, this analysis used Cohen's d — the difference between group means divided by the pooled standard deviation — as the ranking metric. Raw mean differences are scale-dependent: a feature measured in large units (e.g. area, in mm<sup>2</sup>) will always dominate a raw-difference ranking over a feature on a small or normalised scale (e.g. symmetry, a ratio near 0–1), regardless of how well it actually separates the two diagnostic classes. Cohen's d standardises for this by expressing the difference in units of pooled standard deviation, making every feature directly and fairly comparable. The top five features by absolute Cohen's d were retained for visualisation and interpretation.

## 6. Results

The dataset contained no missing values across all 569 samples and 30 features. The class distribution was 357 benign (62.7%) and 212 malignant (37.3%), indicating a moderately imbalanced dataset. Ranking features by standardized effect size revealed clear and consistent separation between diagnostic classes, with malignant samples showing higher mean values for every one of the top five features (all  $|d| > 2.2$ , conventionally interpreted as a very large effect).

**Table 1.** Top five features ranked by standardized effect size (Cohen's d).

| Rank | Feature Name         | Malignant Mean | Benign Mean | Cohen's d |
|------|----------------------|----------------|-------------|-----------|
| 1    | worst concave points | 0.1822         | 0.0744      | 2.69      |
| 2    | worst perimeter      | 141.37         | 87.01       | 2.60      |
| 3    | mean concave points  | 0.0880         | 0.0257      | 2.55      |
| 4    | worst radius         | 21.13          | 13.38       | 2.54      |
| 5    | mean perimeter       | 115.37         | 78.08       | 2.29      |



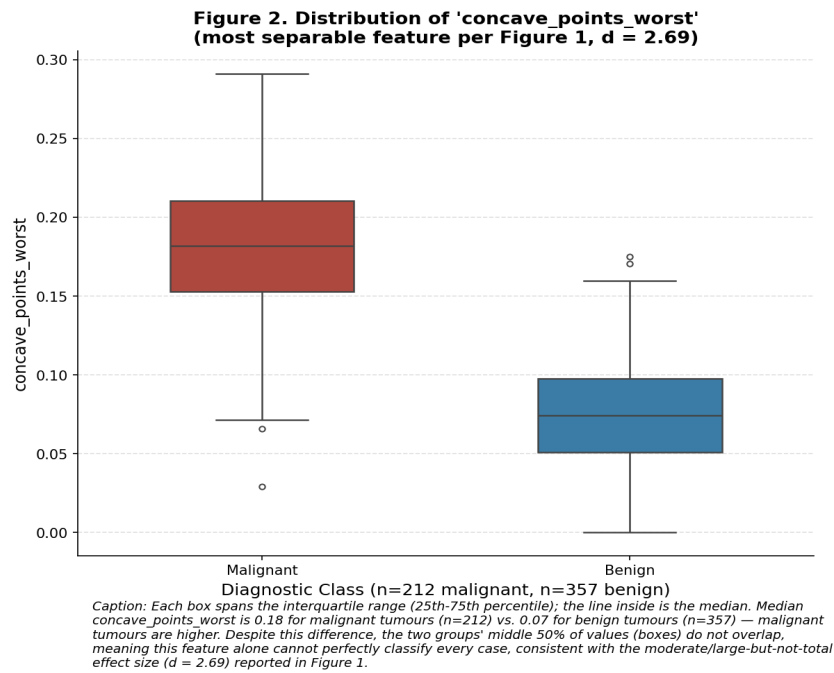


Figure 2. Distribution of worst concave points across malignant and benign diagnostic classes, the single feature with the largest effect size ( $d = 2.69$ ). The embedded caption reports the median, sample sizes, and interquartile overlap for this specific feature.

## 7. Discussion

The results demonstrate that features related to nuclear boundary concavity, perimeter, and size — specifically concave points, perimeter, and radius measurements — are the strongest discriminators between malignant and benign breast tumour samples, each showing a very large standardized effect size ( $d > 2.2$ ). This is consistent with established cancer biology: malignant cells characteristically exhibit irregular, concave nuclear boundaries and nuclear enlargement compared to normal or benign cells (Hanahan and Weinberg, 2011). The dominance of 'worst' (largest value) measurements over mean values for concave points and perimeter suggests that the most extreme cell nuclei within a sample — rather than the average — may carry the strongest diagnostic signal, a finding that aligns with the known heterogeneity of malignant tumour tissue.

The boxplot for worst concave points (Figure 2) further illustrates this separation: the interquartile range of malignant samples is markedly higher than that of benign samples. While the two distributions are largely distinct, a small number of individual benign samples do reach into the lower range of malignant values, meaning this single feature alone cannot perfectly classify every case — a limitation the effect size in Figure 1 ( $d = 2.69$ , very large but not infinite separation) already anticipates. These observations are consistent with the findings of Mangasarian et al. (1995), who identified nuclear boundary and size features as important predictors in their original linear programming classifier.

This study has several limitations. Although ranking features by Cohen's  $d$  addresses the scale-dependence of raw mean-difference comparisons, effect size alone does not establish statistical significance; a more rigorous approach would supplement it with formal hypothesis testing (e.g. Welch's  $t$ -test or the Mann-Whitney  $U$  test) and correction for multiple comparisons across all 30 features. Additionally, the dataset originates from a single institution and may not generalise to all clinical populations. Future work could extend this exploratory analysis by applying supervised

machine learning classifiers — such as logistic regression or random forest — to quantify predictive performance and identify feature interactions that simple group comparison cannot capture.

## 8. Conclusion

This project successfully met all five stated objectives: the dataset was retrieved, cleaned, and analysed to identify the top five most discriminatory cellular features, which were visualised and interpreted in the context of cancer cell biology. Ranking features by standardized effect size (Cohen's  $d$ ), rather than raw mean difference, confirmed that nuclear boundary concavity, perimeter, and size measurements — particularly worst concave points — provide very large, statistically meaningful separation between malignant and benign breast tumour samples. These findings highlight the potential of simple, reproducible, and appropriately standardized statistical workflows to extract meaningful biological insight from open-access diagnostic datasets, and lay the groundwork for more advanced predictive modelling in future work.

## References

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## GITHUB Repository Link

<https://github.com/slrz-pr/Exploratory-Analysis-of-Breast-Cancer-Diagnostic-Features-CAPSTONEPROJECT/tree/main>